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clc;
clear;
close all;

% Parameters
Am = 1;
Ac = 1;
fm = 5;
fc = 50;
fs = 20 * fc;

% Modulation index
mu = Am / Ac;

% Carrier power
Pc = Ac^2 / 2;

% Upper and lower sideband power
Pusb = (mu^2 * Pc) / 4;
Plsb = (mu^2 * Pc) / 4;

% Total sideband power
Psb = Pusb + Plsb;

% Total transmitted power
Pt = Pc + Psb;

% Display results
fprintf('Modulation Index = %.2f\n', mu);
fprintf('Carrier Power = %.3f W\n', Pc);
fprintf('Upper Sideband Power = %.3f W\n', Pusb);
fprintf('Lower Sideband Power = %.3f W\n', Plsb);
fprintf('Total Sideband Power = %.3f W\n', Psb);
fprintf('Total Transmitted Power = %.3f W\n', Pt);

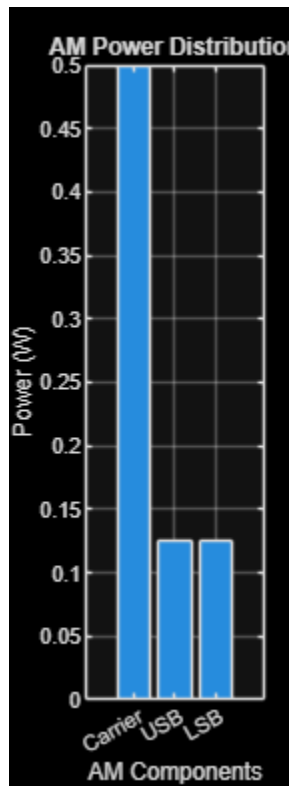
% Power distribution plot
figure('Name', 'AM Power Distribution', 'NumberTitle', 'off');

bar([Pc Pusb Plsb]);
set(gca, 'XTickLabel', {'Carrier', 'USB', 'LSB'});
xlabel('AM Components');
ylabel('Power (W)');
title('AM Power Distribution');
grid on;

Modulation Index = 1.00
Carrier Power = 0.500 W
Upper Sideband Power = 0.125 W
Lower Sideband Power = 0.125 W
Total Sideband Power = 0.250 W
Total Transmitted Power = 0.750 W

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